

EUROPEAN ECONOMICS ANALYST

Germany—Autoland Under Pressure

- Germany's auto industry is the heart of its manufacturing base. It directly accounts for 4% of total German value added and 2% of employment—three to four times the share of peers—plus another 2% of value added indirectly via domestic supply chains. Since peaking in 2017/18, the sector has endured a prolonged downturn, with orders, production, sales, and employment all declining. This slowdown alone has reduced GDP growth on average by 0.3pp annually in recent years compared with the boom years after the financial crisis, contributing roughly a fifth to Germany's broader economic weakening.
- The slowdown is almost equally due to global and domestic factors. First, weak demand hit production between 2018 and 2021, initially a German idiosyncrasy as post-GFC outperformance faded. Second, global supply chain disruptions weighed heavily on the global industry in 2021–2023, but affected Germany relatively more than EU peers. Third, since 2024 German producers have suffered from subdued demand, particularly in Europe, alongside intensifying competition from China.
- Although aggregate auto sales across German export markets have rebounded since 2024, German production and sales have underperformed—shedding market share notably in China and other non-EU markets such as the UK, Japan, and the US. Chinese competition is the key headwind. China's price competitiveness has improved sharply since 2022, and German market share losses closely track rising Chinese market penetration. US exports held firm until last year, but new tariffs have pushed motor vehicle exports roughly 25% below pre-tariff levels.
- The EV transition is advancing fast, but at a cost. Battery electric vehicles and plug-in hybrids now make up almost 40% of German production. Yet electric engines require less labour, and a high share of batteries is imported—squeezing domestic employment and value added.
- Taken together, continued subdued demand in Europe, rapidly rising competition from Chinese producers in their home market (and particularly in export markets), US tariffs and the shift to electric vehicles make for a challenging environment for the German auto industry. In a scenario where these headwinds weigh on the domestic share of value added and reduce German market share across markets, value added in the Autos industry could fall by 8% by 2030, creating a 0.1pp annual drag on aggregate growth. With a faster erosion of

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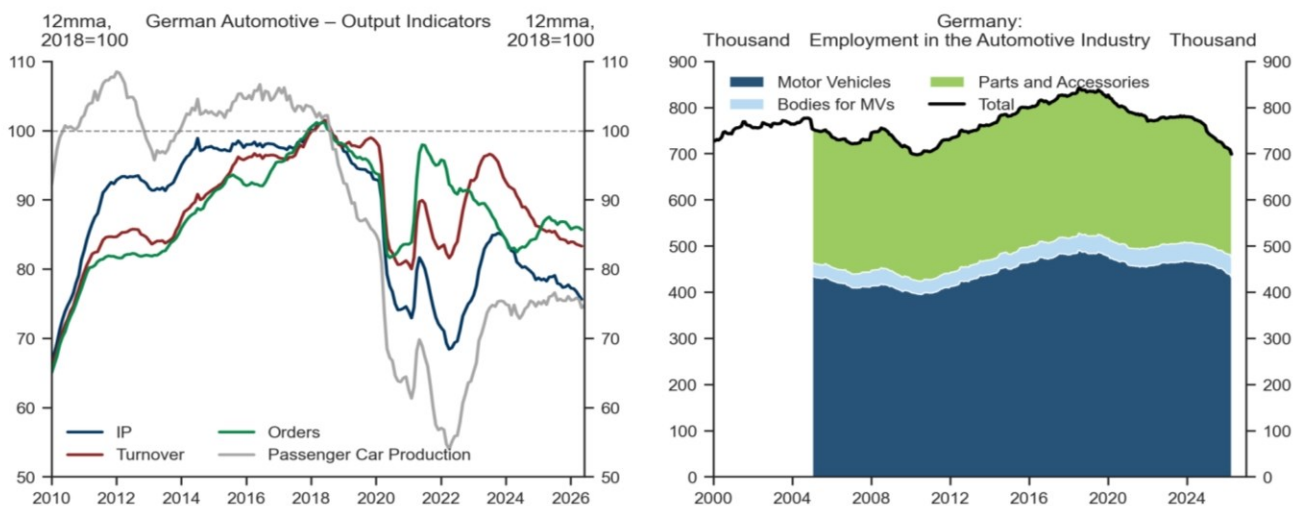
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market shares due to Chinese competition this drag could double, while a build out of the domestic value chain for electric vehicles could induce limited growth over the next years.

Germany—Autoland Under Pressure

Since peaking in 2017/18, the German automotive industry has been in a long downturn ([Exhibit 1](#)). Production has declined by 25% to 30% below its peak, real orders and revenues have declined by about 15%, and employment is also down almost 20%, particularly for suppliers.¹ [Plans](#) by major German car manufacturers to cut more jobs and reduce production capacity in Europe have raised concerns over the impact on the German economy. In this *Analyst* we take stock of the macroeconomic relevance of Germany's automotive industry, the reasons for the recent downturn and the outlook for car manufacturing in Germany.

Exhibit 1: The German Auto Industry Has Been Struggling Since 2017/18



RHS: Employment is measured for all companies with 50 or more employees. Employment at smaller companies accounts for about another 119k employees in the aggregate auto industry.

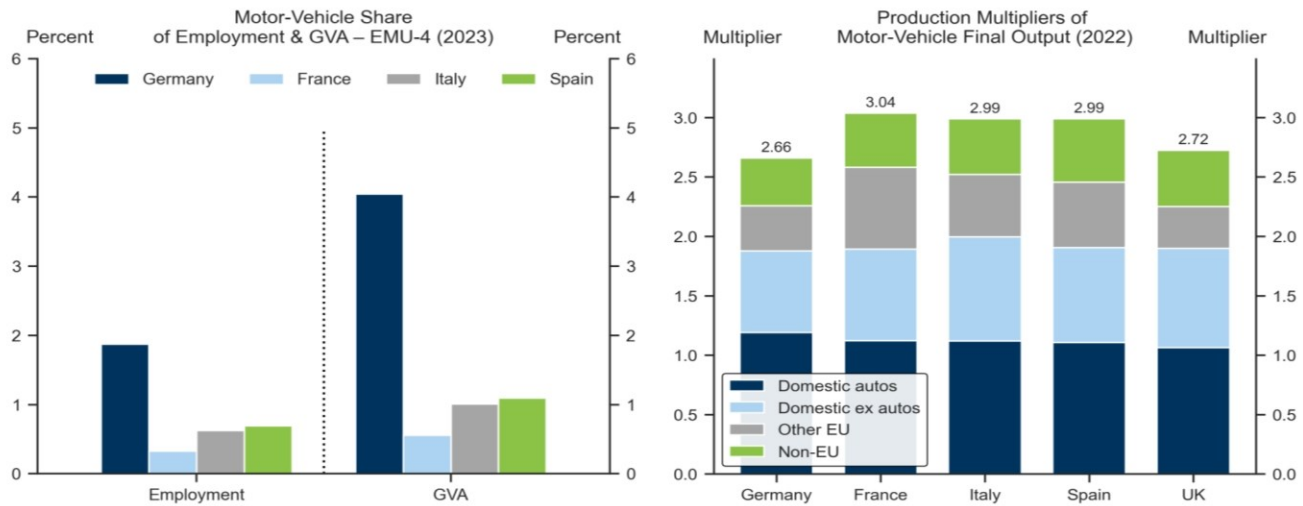
Source: Haver Analytics, Goldman Sachs Global Investment Research

Autoland Germany

Germany's car industry is at the heart of German manufacturing. It directly accounts for 4% of total value added and 2% of total employment, more than three to four times the share in other large European economies ([Exhibit 2](#), left). Through supply chain linkages, the car industry supports another 2% of value added indirectly in other domestic industries ([Exhibit 2](#), left).²

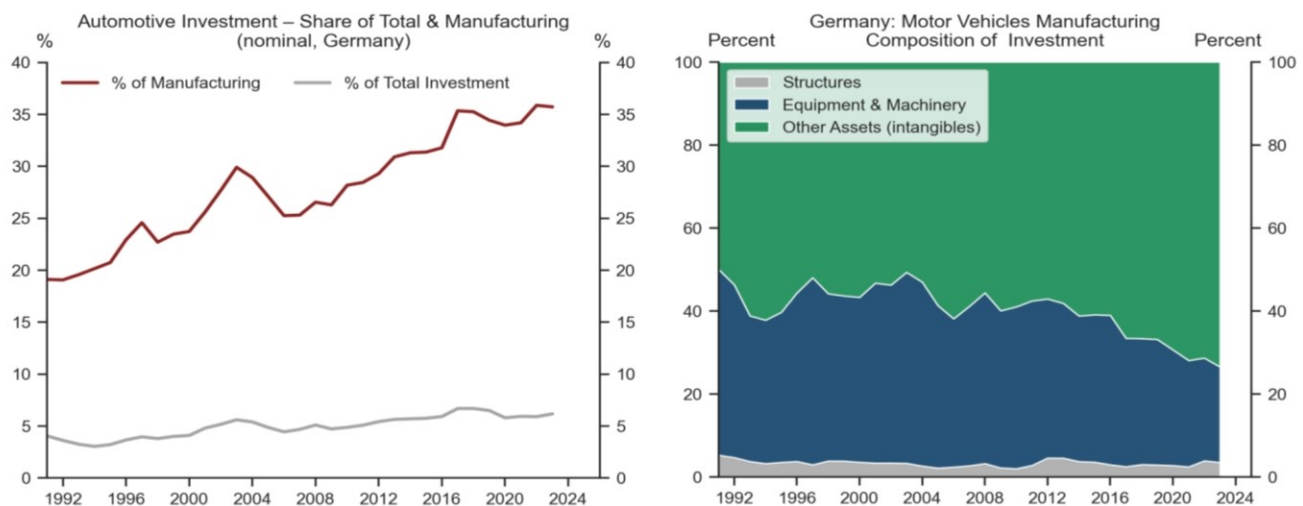
¹ The stronger decline in production relative to real turnover is likely related to a shift in the product mix towards higher value-added cars, particularly while supply chains were disrupted in 2021 to 2023.

² In [Exhibit 2](#) we show the production multiplier of one unit of autos value added. A multiplier of 2.66 means that to produce one unit of autos value added, another 1.66 units of value added of other sectors are required throughout the supply chain. For Germany about 0.2 further units of GVA are generated in the domestic auto industry and 0.6 units in other domestic industries. This implies a ratio of indirect value added in other domestic industries to total value added in the car industry of about 1.5.

Exhibit 2: Germany Stands Out for Its Large Auto Industry

Source: Haver Analytics, Goldman Sachs Global Investment Research, OECD

Apart from its contribution to production, the car industry is also central for business R&D and investment in Germany. Its share in total investment has steadily trended up since reunification, reaching more than 6% of aggregate fixed investment and almost 36% of manufacturing fixed investment in 2023, both substantially more than the respective value-added share of 4% and 20% (Exhibit 3, left). The composition of investment has tilted towards intangibles, which made up more than 70% of investment in recent years, in line with the fact that the auto industry is the largest spender on business R&D in Germany, accounting for between a quarter and a third of spending (Exhibit 3, right).

Exhibit 3: The Automotive Industry Accounts for 6% of Investment, with Spending Tilted to Intangibles

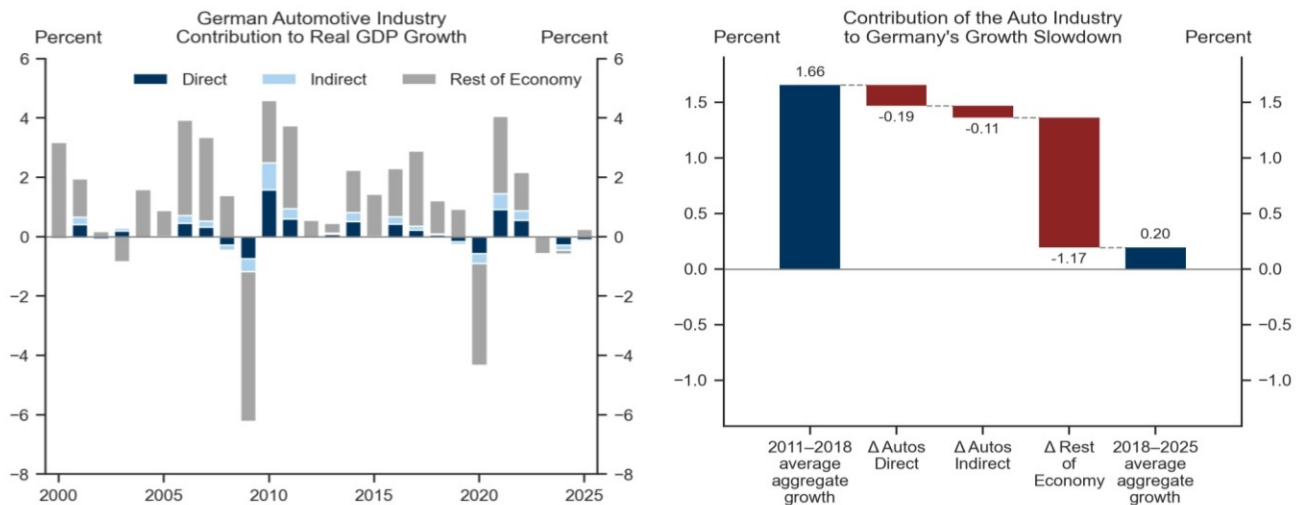
Source: Haver Analytics, Goldman Sachs Global Investment Research

Real value added growth in the auto industry slowed from +5.1% per year between 2011 and 2018 to +0.4% per year between 2018 and 2025 (Exhibit 4, left).³ While value added

³ We choose 2011 as the starting year to avoid the strong rebound after the financial crisis. As value added

growth slowed down significantly, it performed much better than industrial production or real turnover, likely due to a combination of an improving product mix and a higher share of added services that are not captured by the production numbers. We estimate that the slowdown in auto industry growth has lowered aggregate growth by 0.3pp through direct and indirect effects, thus contributing about a fifth to the slowdown of the German economy (Exhibit 4, right).⁴

Exhibit 4: Slower Growth in the Automotive Industry Reduced Aggregate GDP Growth by 0.3pp annually in 2018-2025 vs. 2011-2018



Source: Haver Analytics, Goldman Sachs Global Investment Research, OECD

Three Phases of a Slowdown: Demand Weakness, Supply Disruptions, Chinese Competition

German car production slumped in three distinct phases. In the first phase between 2018 and 2021, lower demand weighed on German production (Exhibit 5, left). While the German car industry experienced stronger demand than its European peers until 2018, demand weakened relative to its peers pre-Covid, as diesel sales plummeted in the wake of the diesel scandal and stricter environmental regulations in Europe. Subdued demand during Covid hit Germany in a broadly similar way to other European economies.

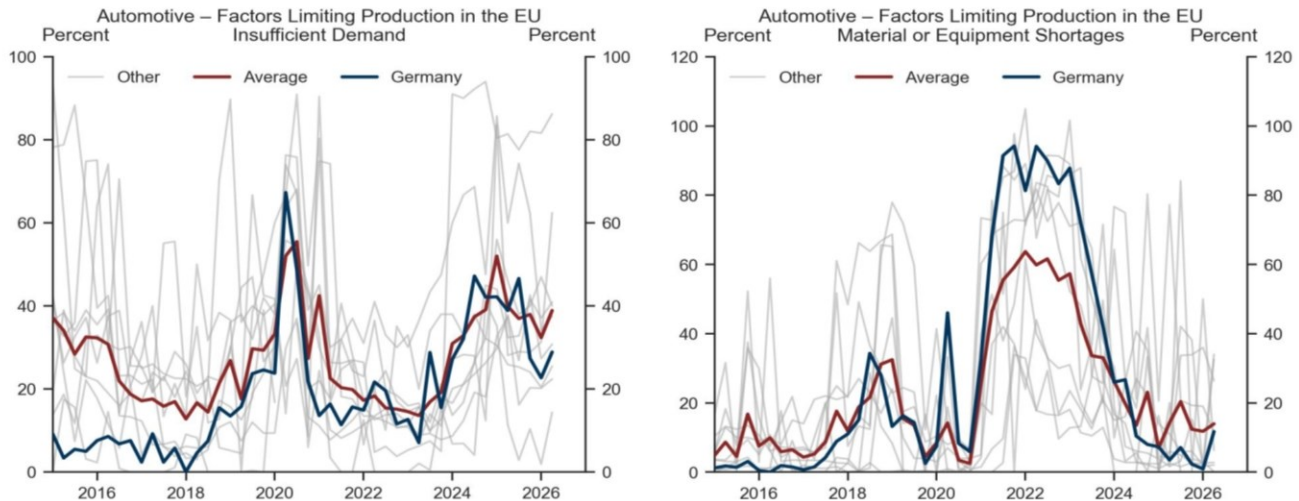
The second phase, during 2021 to 2023, was characterised by global supply chain disruptions, particularly for semiconductors (Exhibit 5, right). While supply shortages weighed on production across the EU, the German car industry was affected relatively more strongly, although it then saw a stronger rebound in 2023.

Finally, after recovering from supply chain disruptions, the third phase since 2024 has been characterised by subdued demand, with the German car industry affected similarly to the mean across the EU.

data are only available until 2023, we project value added growth forward based on real turnover in the automotive industry. A projection based on industrial production yields similar results. That said, given the outperformance of value added relative to both turnover and industrial production, these estimates should be viewed as upper bounds on the drag from the auto industry.

⁴ The growth contribution was about 0.3pp per year in 2011 to 2018 and was only slightly above flat in 2018 to 2025.

Exhibit 5: Low Demand Weighed On German Production Between 2019 and 2021 and Since 2024, Supply Chain Disruptions Were Material in 2021 to 2023

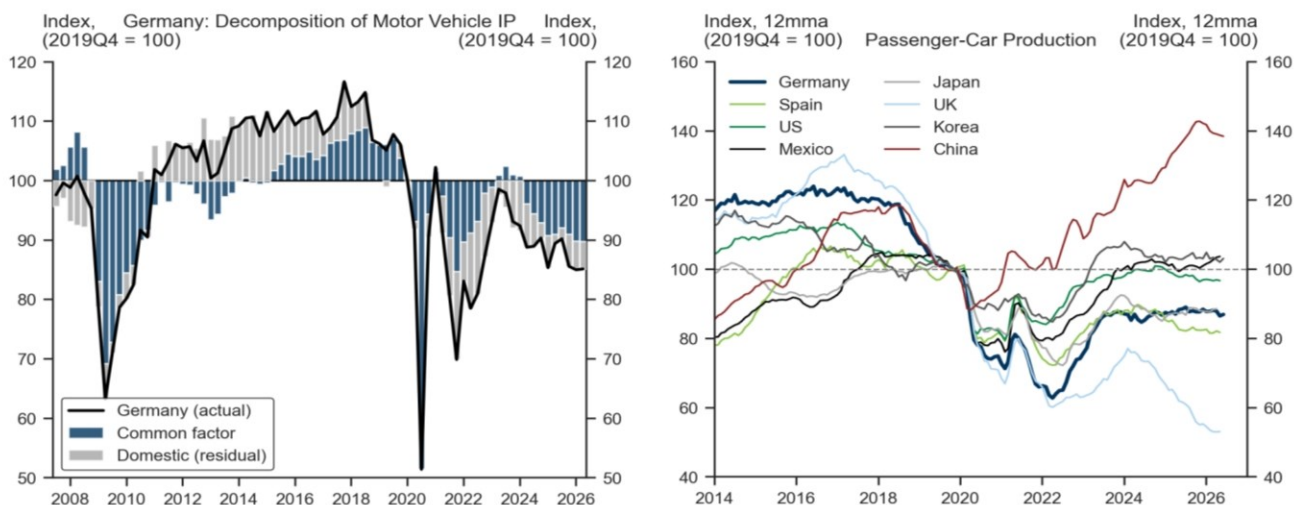


Source: Haver Analytics, Goldman Sachs Global Investment Research

Decomposing German automotive production into a global factor (the first principal component of 27 countries' auto production) and a domestic factor also indicates that the slump in 2018 was mostly a German idiosyncrasy as German outperformance after the financial crisis faded ([Exhibit 6, left](#)). Since 2019, most of the decline in German production can be attributed to the global factor, in particular the slump since 2024.

This factor decomposition based on real industrial production is broadly consistent with the number of passenger cars produced ([Exhibit 6, right](#)). Germany underperformed in 2018 and during the supply chain crisis and has moved broadly in line with its peers since then. The major exception is Chinese car production, which has surged by 25% since 2023, while production in other economies flatlined.

Exhibit 6: The German Auto Industry Underperformed in 2018 and during the Supply Chain Disruptions but Otherwise Moved Broadly with the Global Cycle



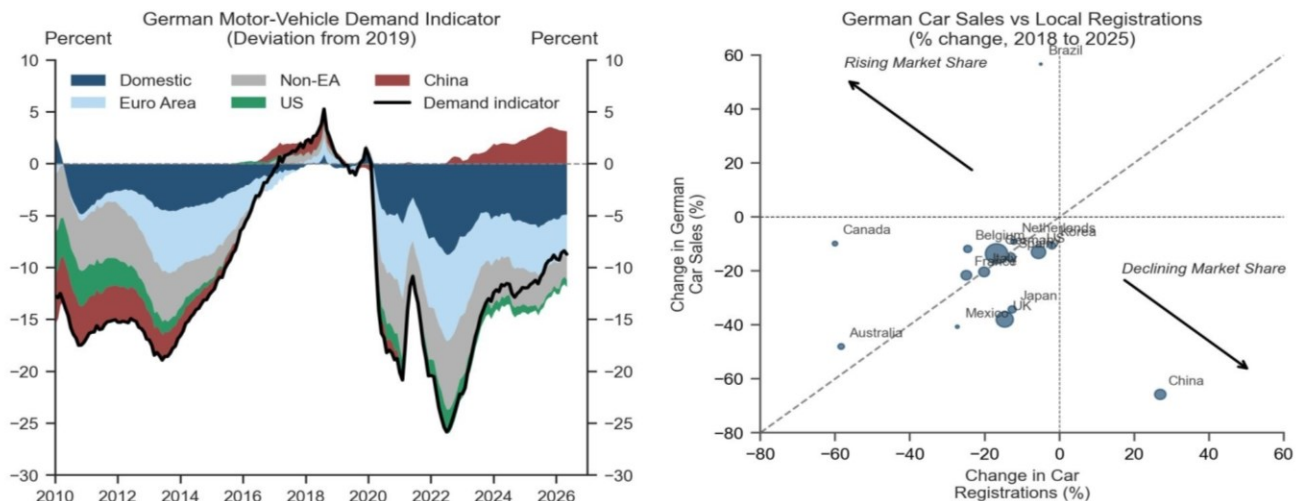
Source: Haver Analytics, Goldman Sachs Global Investment Research

To further analyse the drivers of the demand slowdown we construct a demand

indicator for the German auto industry based on motor vehicle registrations across the auto industry's major sales markets.⁵ Between 2012 and 2022 sales volumes closely track this demand indicator, which rose until 2018, then declined by 5% pre-Covid and by another 20%-25% until mid-2022 ([Exhibit 7, left](#)). This suggests that the pre-Covid downturn was mostly driven by lower car sales across the specific markets of the German car industry.

Since 2023 demand has recovered somewhat propelled by stronger demand in China and outside of the Euro area, while domestic and Euro area demand have remained subdued. That said, production and sales declined, which suggests that German producers have lost market share. This loss in market share is particularly visible in China, but also in Japan and the UK, while sales of German-produced cars moved broadly in line with car registrations or even outperformed them slightly in most other markets ([Exhibit 7, right](#)).⁶ While German exports to the US held up relatively well until 2025, motor vehicles and parts exports declined after the introduction of US tariffs last year and have now settled around 25% below the pre-tariff level in value terms.

Exhibit 7: Sales of German-Produced Cars Broadly Moved with Demand until 2022 but Recently Underperformed, Particularly in China



Source: Haver Analytics, Goldman Sachs Global Investment Research, World Bank

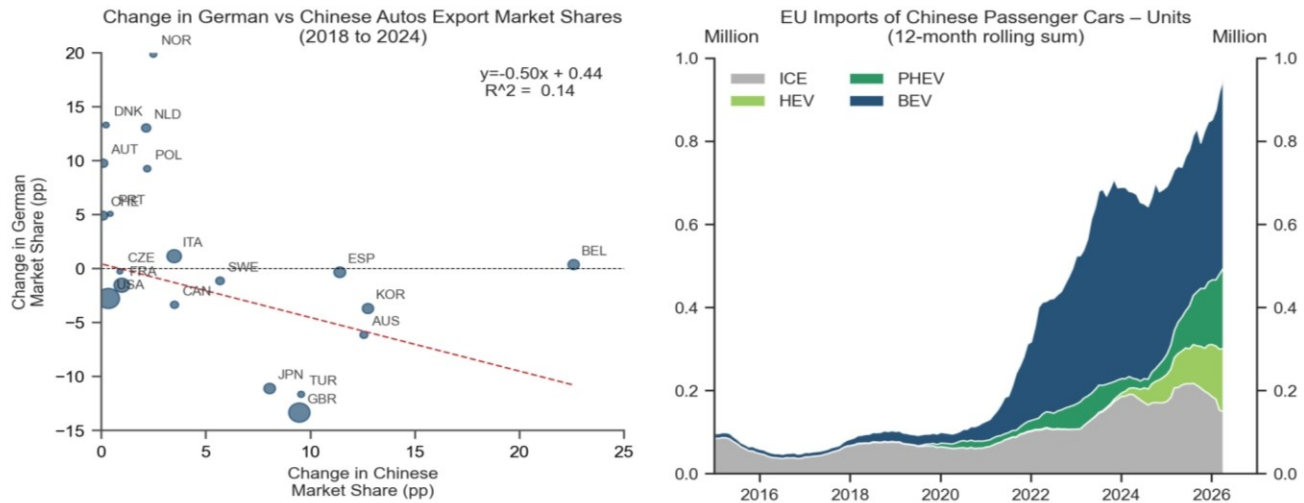
The underperformance of German exports to the Chinese market, particularly since 2023, seems related to increasing competitive pressure from Chinese manufacturers and a shift to electric vehicles (EVs) in the Chinese market. Furthermore, Chinese exports are increasingly putting pressure on German exports in third markets, where German market share losses are correlated with Chinese market share gains ([Exhibit 8, left](#)). While export market share losses have so far mostly occurred outside of the EU, Chinese car exports are increasingly directed to the European market ([Exhibit 8, right](#)). The introduction of EU tariffs on Chinese EVs in 2024 slowed EV imports from China. That said, import growth resumed quickly as imports adjusted towards hybrids, which are exempted from

⁵ To construct the indicator we index the number of new motor vehicle registrations for major sales markets of the German car industry to 100 in 2019 and take a weighted average, with the weights corresponding to the sales share of the German car industry in 2019.

⁶ About 25% of the decline in German car production can be accounted for by lower exports to the UK and the EU respectively, 20% by lower exports to China, 5% by lower exports to the US and 15% by lower domestic sales.

tariffs.

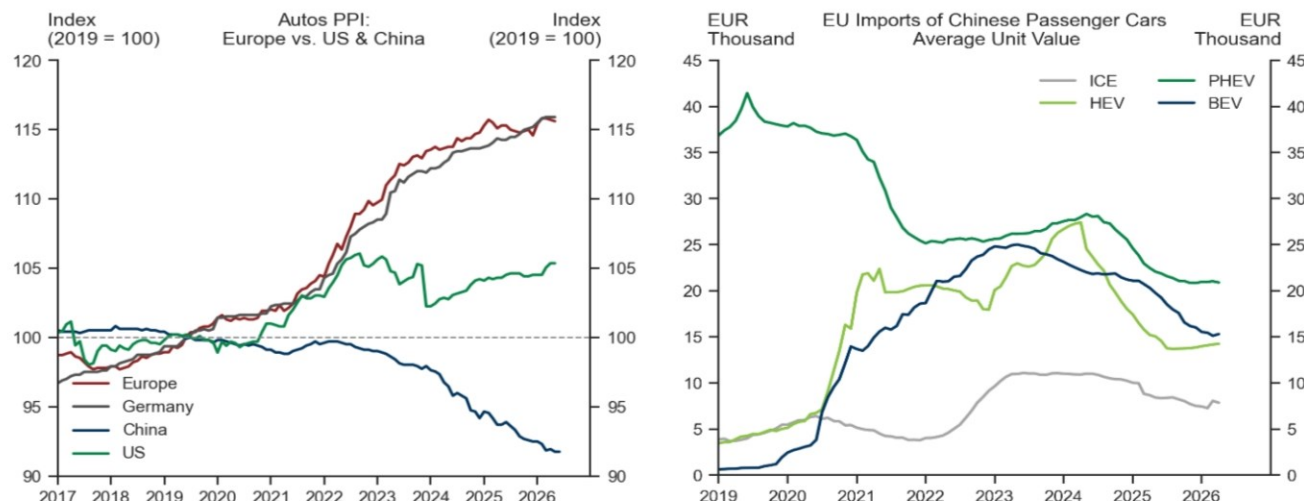
Exhibit 8: Chinese Exports Are Putting Pressure on the German Car Industry



Source: Haver Analytics, World Bank, Goldman Sachs Global Investment Research

Market share gains of Chinese manufacturers are supported by their price competitiveness, which continues to be supported by substantial subsidies. Their cost position has improved in recent years, particularly since 2022, with PPIs declining by more than 20% relative to European manufacturers ([Exhibit 9, left](#)).⁷ The average price of European car imports from China has also declined meaningfully, particularly for electric vehicles and hybrids, contributing to rising market penetration ([Exhibit 9, right](#)). The continued surge of cheap Chinese car exports is thus likely to continue to put pressure on the German car industry.

Exhibit 9: Chinese Auto Manufacturing is Highly Price Competitive



Source: Haver Analytics, Goldman Sachs Global Investment Research

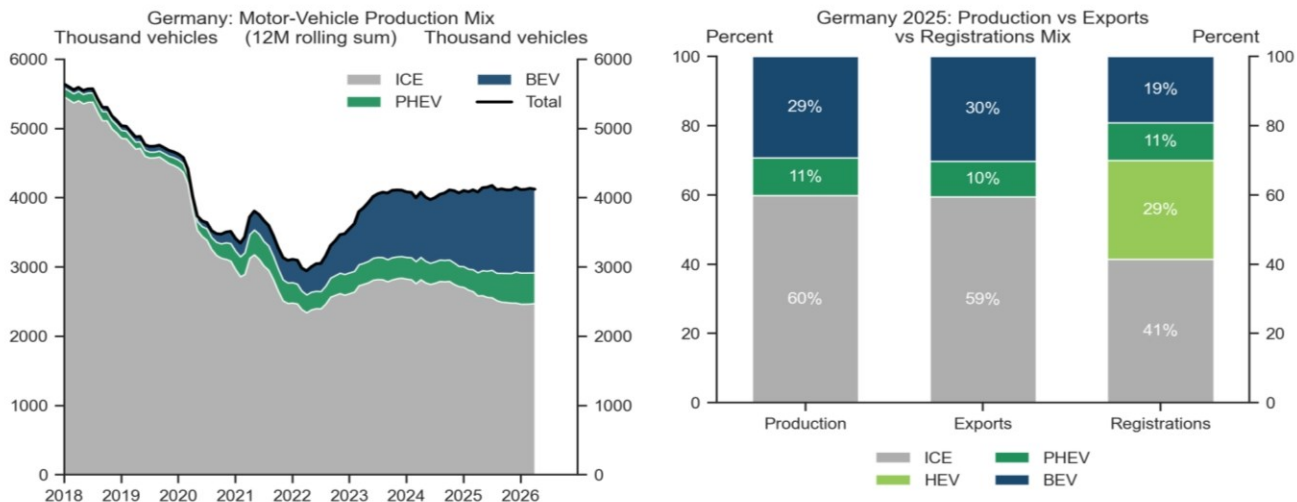
German Production Shifts to EVs, Putting Pressure on the Domestic ICE

⁷ The substantial improvement in the relative cost position also remains when adjusting for currency effects.

Value Chain

In addition to the increase in competition from China, the shift to EVs is changing the structure of the German car industry. Our equity analysts expect the share of BEVs (battery electric vehicles) and PHEVs (plug in hybrid vehicles) in global car sales to increase to almost 40% by 2030, further rising to about 70% in 2040. Since 2022 the mix of German car production has shifted towards EVs (Exhibit 10, left). Last year BEVs and PHEVs accounted for around 40% of production and exports (Exhibit 10, left), about 10pp higher than the share of EV registrations.⁸ Non-plugin hybrids (HEVs), which account for another 30% of registrations, are currently not produced in Germany but imported.

Exhibit 10: Production and Registrations are Shifting Towards EVs



Source: Haver Analytics, Goldman Sachs Global Investment Research

The shift to EVs is putting pressure on labour demand and car suppliers, particularly those focused on the combustion powertrain. Electric engines are less complex than combustion engines and the associated transmission system, with estimates suggesting that electric engines require only two-thirds of the labour input of combustion engines. Employment in the auto industry has already declined by about 150,000 since its peak in 2019, with suppliers bearing the majority of those losses. A study by the ifo institute indicated that around 250,000 jobs that depend on combustion engines would be affected by the transition to EVs between 2019 and 2030. While overall job losses could be cushioned somewhat by rising labour demand for EV components such as batteries, this hinges on a material share of production occurring in Germany.⁹ Reallocation of labour towards the growing defence industry could also cushion job losses in the auto sector.

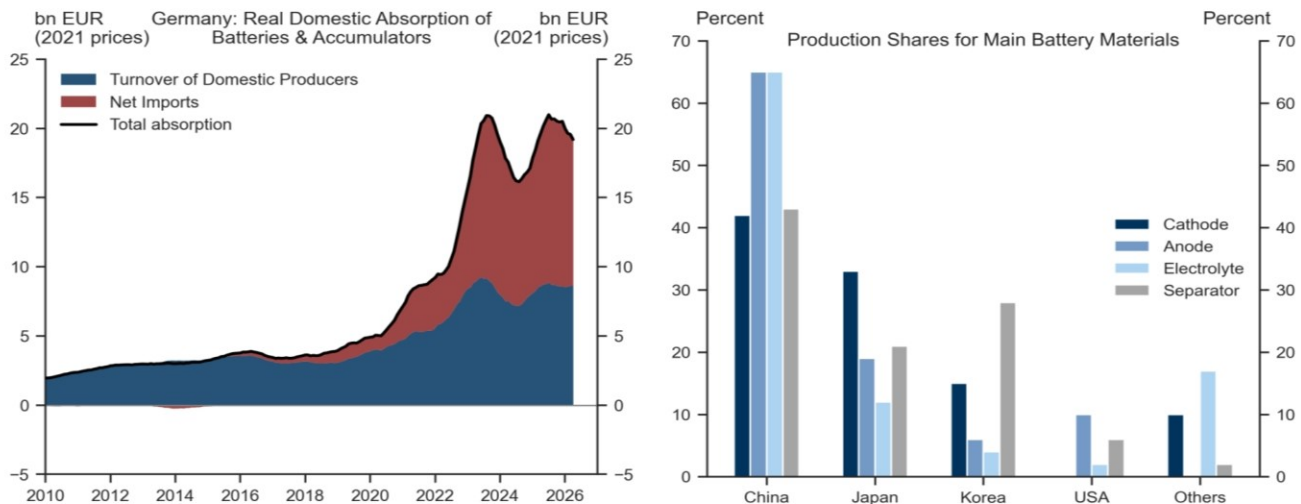
That said, while battery production has been rising in Germany, a substantial share of domestic absorption is currently met by imports (Exhibit 11, left) and production of critical inputs required for battery production is concentrated outside of Europe (Exhibit 11, right), lowering the domestic value-added share of EV production. Estimates put the

⁸ The registration share has since risen further to 34% in June 2026 on a 12-month rolling basis.

⁹ Employment gains for the production of EV components could offset about half of the job losses due to lower ICE production in the auto industry, and rising labour demand for energy infrastructure could provide a further offset.

relative European value added of a BEV produced in Europe at 75%, compared with 90% for a European-produced combustion engine car.

Exhibit 11: The Demand for Batteries Has Risen Substantially, But More Than Half of Demand is Met by Imports

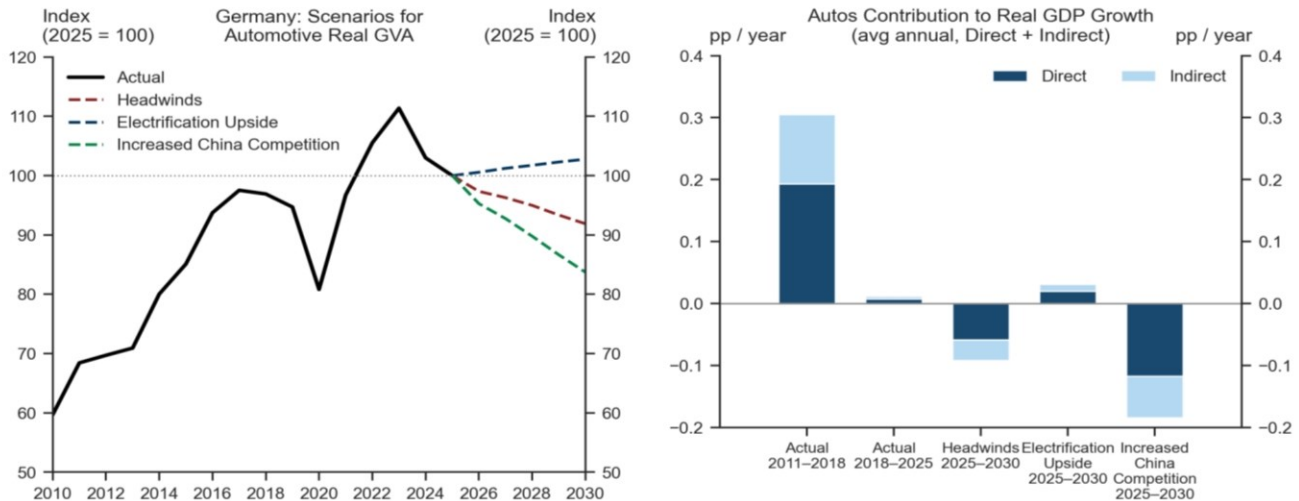


Source: Haver Analytics, Refinitiv Eikon, US Department of Commerce, Goldman Sachs Global Investment Research

Taken together, continued subdued demand in Europe, rapidly rising competition from Chinese producers in their home market (and particularly in export markets), US tariffs and the shift to electric vehicles make for a challenging environment for the German auto industry.

Orders and European demand have stabilised, suggesting that the decline in auto production may have bottomed out for now. However, structural headwinds are likely to limit any recovery and pose risks to the downside. Based on our autos analysts' longer-term projections for car sales by market, we estimate that the German autos industry could shrink by more than 8% by 2030 in a scenario where German exporters lose further market share in China, exports fail to recover in the US, market shares in Europe decline slightly, and further electrification reduces the domestic value-added share of cars ([Exhibit 12](#), left). In this scenario, the drag from autos on aggregate growth would be about -0.1pp per year ([Exhibit 12](#), right).

Exhibit 12: While Headwinds Linger, Capturing a Larger Share of the EV Value Chain is an Opportunity for the German Autos Industry



Source: Haver Analytics, Goldman Sachs Global Investment Research

In a downside scenario where increased competition from China more strongly weighs on German market share in Europe and abroad, autos value added could decline by 16% between 2025 and 2030, resulting in a 0.2pp drag on GDP growth.

That said, while risks for the industry are tilted to the downside, a build-out of the domestic EV value chain could be an opportunity for the German auto industry. In a scenario where production shifts towards EVs more quickly but a larger share of the EV value added is captured domestically and market shares remain at their 2025 levels, value added could increase by 5%, resulting in a tailwind to growth of 0.06pp.

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